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# A Multi-Modal Architecture for Document Forgery Detection

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***Abstract*—Document fraud has evolved into a technically sophisticated threat, involving high-resolution print–scan recapture, selective region manipulation, and adversarial rewriting. This work presents a multimodal deep architecture for robust forgery detection that integrates visual texture cues, structural edge patterns, and textual semantics. The system employs an MobileNet-based visual backbone augmented with CBAM and a lightweight ASPP module to capture fine-grained manipulation artifacts. Parallel pathways model structural inconsistencies via edge analysis and linguistic deviations through OCR embeddings, targeting splicing and font substitution attacks respectively. The three modalities are fused through a normalized integration block and classified using an MLP head. Experiments conducted on a 10k-image IDNet-Estonia-derived corpus demonstrate an accuracy of 97.8 and an AUC of 0.9992 across seven fraud categories. Qualitative visualizations further confirm the model's ability to localize manipulation artifacts, significantly outperforming unimodal baselines in complex attack scenarios.**

***Index Terms*—Document Forgery Detection, Multimodal Deep Learning, MobileNetV2, Optical Character Recognition (OCR), Edge Detection, CBAM Attention Module, Atrous Spatial Pyramid Pooling (ASPP), Image Tampering Detection.**

## I. Introduction

Document forgery represents an escalating cybersecurity and forensic challenge as digital and physical authentication systems increasingly converge. Traditional countermeasures—such as watermarking, optical security features, and template matching—have become insufficient against sophisticated forgeries that exploit high-resolution print–scan recaptures, selective manipulations, and semantic text alterations [9] . Recent studies have underscored the limitations of unimodal systems that focus solely on pixel-level or OCR-based analysis[1]. Image-only detectors struggle with high-quality print–scan re-encodings and local inpainting attacks [8], while text-only models fail to capture texture or edge inconsistencies introduced during forgery [4].

In this work, we propose a multimodal deep architecture that unifies three orthogonal evidence streams—visual texture, structural edge features, and OCR-based semantic embeddings—into a cohesive representation for robust document forgery detection. The visual backbone employs MobileNet enhanced with a Convolutional Block Attention Module (CBAM) and a lightweight Atrous Spatial Pyramid Pooling (ASPP) module to capture fine-grained manipulation artifacts across scales. In parallel, a dedicated edge pathway models geometric inconsistencies indicative of copy–move or morphing operations [6], while an OCR embedding stream encodes linguistic anomalies and font inconsistencies[7] .Our contributions are threefold:

We introduce a unified multimodal deep architecture integrating visual, structural, and textual modalities for comprehensive document forgery detection.

We design a normalized feature fusion framework that effectively balances multi-source embeddings without introducing modality bias.

We perform extensive benchmark evaluations and ablation studies demonstrating superior robustness and interpretability through attention visualizations and confusion analyses.

*These authors made equal contributions.


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Fig. 1: Face Morphing Example




Fig. 2: Genuine Document

## II. Literature Survey

Recent research in document forgery detection (2022–2025) shows a strong shift toward CNN-based hybrid models, often combined with OCR, SIFT, LBP, GANs, and transformer architectures. Many works focus on region-based verification, signature forgery, passport/ID tampering, and low-quality image enhancement are common targets in recent work. Techniques such as multi-task CNN–RNN pipelines and OCR-guided verification consistently achieve high accuracy (92–98%).

## III. Dataset

### A. Dataset Origin

The dataset used in this study is derived from the Estonia subset of the IDNet benchmark dataset (IDNet–EST), a publicly available corpus designed for document forgery detection and identity manipulation researches [15]. Each document instance in IDNet comprises one genuine sample and six systematically generated forged variants, corresponding to distinct fraud categories: copy-and-move, inpaint-and-rewrite, face morphing, portrait substitution, direct text-field replacement, and combined forgery patterns.

Each document instance is organized into separate directories, where each folder represents a specific forgery category. This structured organization enables precise segmentation of forgery types and facilitates controlled experimental evaluation for both classification and verification tasks. Notably, both the facial images and textual fields within the ID cards are entirely synthetically generated, ensuring data privacy and controlled forgery simulation [15].

### B. Class Definitions

The dataset comprises seven mutually exclusive classes:

- positive
- fraud1_copy_move
- fraud2_face_morph
- fraud3_face_replace
- fraud4_combined
- fraud5_inpaint_rewrite
- fraud6_crop_replace

### C. Data Splitting Strategy

The dataset is partitioned into training, validation, and testing subsets using stratified sampling to preserve class distribution consistency. The split ratio is defined as follows:

- 70% for training,
- 15% for validation,
- 15% for testing.

Stratified sampling is implemented using the *scikit-learn* library to ensure balanced representation of all forgery categories across each subset.

## IV. Methodology

### A. System Overview

The proposed forgery detection framework integrates three complementary information streams—visual appearance, structural boundaries, and textual semantics—to capture a wide spectrum of manipulation cues commonly found in identity documents. Each modality is processed by an independent branch designed to extract features relevant to its domain. The resulting embeddings are fused into a unified representation and classified using a multilayer perceptron. This design enables the model to jointly reason about pixel-level noise patterns, geometric inconsistencies, and OCR-level content deviations.

### B. Preprocessing

All images are resized to $256 \times 256$ and normalized using ImageNet statistics. During training, random flips, small-angle rotations, and color jitter are applied to simulate distortions introduced by various acquisition devices.

Let $I \in \mathbb{R}^{256\times256\times3}$ denote the normalized image. A Canny detector extracts the structural boundary map

$$E = \text{Canny}(I),$$

which highlights edges susceptible to geometric inconsistencies caused by splicing or copy–move operations.

OCR text is extracted from the image and converted into a sequence of fixed length 64:

$$T = [t_1, t_2, \ldots, t_{64}],$$

ensuring uniformity for downstream text processing.

### C. Visual Branch

The visual branch is tasked with identifying pixel-level irregularities such as abnormal textures, blending artifacts, or color inconsistencies. MobileNet V2 is used as the core feature extractor due to its strong performance across image classification tasks.


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TABLE I: Comparative summary of recent document forgery detection literature (2023–2025).

| Author(s) | Focus | Key Points | Techniques |
|---|---|---|---|
| S. Lacca et al. (2025) | Hybrid multi-task CNN | CNN-based multi-task learning integrating text and texture; 96.8% accuracy. | CNN, OCR, Expert Labels |
| M. Kurvey et al. (2024) | Veripass passport detection | CNN+SIFT+OCR based verification achieving 95.2%. | CNN, OCR, SIFT |
| R. Sukhija et al. (2025) | Forgery survey | Analysis of LBP, attention mechanisms, and hybrid models. | CNN, LBP, OCR |
| Y.Y. Bae et al. (2025) | Edge-focused DL | Edge-enhanced CNN and adaptive OCR; 97.3%. | Edge CNN, OCR |
| S. Parveen et al. (2024) | Passport security | Faster R-CNN + OCR pipeline for air-security; 98.1%. | Faster R-CNN, YOLOv5, OCR |
| P. Tyagi et al. (2024) | Writer identification | Unsupervised CNN + OCR for handwriting forgery; 94.5%. | CNN, OCR, Clustering |
| R. Murugan et al. (2025) | OCR for degraded docs | CNN–RNN–Transformer hybrid; 93.9%. | CNN, RNN, Transformer, OCR |
| B.M. Tornés et al. (2023) | Receipt forgery dataset | CNN on Find-It-Again dataset; 92.4%. | CNN, OCR Dataset |
| M.S. Arul et al. (2025) | Fake ID detection | CNN + OCR microtext consistency; 95.5%. | CNN, OCR |
| V.J.L.P. Devi et al. (2025) | PAN forgery detection | CNN ensemble + OCR consistency; 94.2%. | CNN Ensemble, OCR |
| R. Bhardwaj et al. (2024) | Historical OCR | CNN + generative preprocessing; 91.8%. | CNN, OCR, GenAI |
| C.K. Sonkar et al. (2025) | Handwriting survey | CNN-RNN hybrid writer verification. | CNN, RNN, OCR |
| V. Sharma et al. (2025) | Handwriting review | CRNN + OCR pipeline. | CNN, CRNN, OCR |
| D.M. Nogueira et al. (2025) | Forgery overview | CNN, GAN, OCR hybrid analysis. | CNN, GAN, OCR |

A Convolutional Block Attention Module (CBAM) is integrated to strengthen the network's ability to focus on visually relevant regions. CBAM refines the intermediate representation by applying channel and spatial attention successively, enhancing sensitivity to subtle manipulation traces. As each channel of a feature map is considered as a feature detector, channel attention focuses on 'what' is meaningful given an input image [13].

To capture multi-scale artifacts—often present at both fine (digit-level) and coarse (full-region) granularities—an Atrous Spatial Pyramid Pooling (ASPP) module is employed. ASPP aggregates information across multiple receptive fields, enabling the network to detect both local and global inconsistencies[27].

A global average pooling layer compresses the final tensor into a 1280-dimensional visual embedding:

$$v_{\text{img}} \in \mathbb{R}^{1280}.$$

### D. Edge Branch

Structural deformation is a common outcome of tampering processes such as region replacement, boundary erasing, and text overwriting. The edge branch focuses on these geometric cues using the edge map $E$.

A lightweight convolutional network processes $E$ through a $7 \times 7$ convolution with stride 2 to capture coarse structural patterns, followed by batch normalization, ReLU activation, and spatial dropout for regularization. Global average pooling is then applied, and the output is projected into a compact 32-dimensional embedding:

$$v_{\text{edge}} \in \mathbb{R}^{32}.$$

By operating solely on shape and boundary information, this branch captures geometric inconsistencies that are typically invisible to RGB-only feature extraction, making it particularly effective for low-quality scanned documents.

### E. OCR Branch

Textual manipulation is a prevalent form of document tampering, commonly involving character rewriting, font substitution, spacing irregularities, and number replacement. To capture such cues, an OCR-based branch processes character-level textual information extracted from document images.

OCR output is converted into a fixed-length character sequence, which is embedded into a 64-dimensional vector space. An attention-based aggregation mechanism is then employed to adaptively weight salient character representations. Specifically, a learnable query vector attends over the embedded sequence using a scaled dot-product formulation, producing a weighted summary representation. The aggregated vector is subsequently projected into a compact 32-dimensional embedding:

$$v_{\text{text}} \in \mathbb{R}^{32}.$$

This attention-driven formulation enables the model to emphasize informative character patterns and typographic irregularities while suppressing noisy OCR artifacts, thereby improving robustness under real-world document degradations.

### F. Attention-Based Fusion Module

Let $v_{\text{img}}$, $v_{\text{edge}}$, and $v_{\text{text}}$ denote the embeddings extracted from the visual, edge, and OCR branches, respectively. These


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embeddings are concatenated to form an intermediate multimodal feature vector:

$$v_{\text{cat}} = [v_{\text{img}} \,\|\, v_{\text{edge}} \,\|\, v_{\text{text}}].$$

To enable adaptive modality weighting, an attention-based fusion mechanism is introduced. A lightweight gating network predicts modality importance scores, which are normalized using a softmax function to obtain attention weights. Each modality embedding is then reweighted according to its learned contribution, yielding:

$$v_{\text{fusion}} = [\alpha_{\text{img}} v_{\text{img}} \,\|\, \alpha_{\text{edge}} v_{\text{edge}} \,\|\, \alpha_{\text{text}} v_{\text{text}}],$$

where $\alpha_i$ denotes the learned attention weight for modality $i$.

This attention-guided fusion strategy allows the network to dynamically emphasize the most informative cues depending on the forgery type, leading to improved generalization and robustness across diverse manipulation scenarios.

### G. Classification Head

The fused embedding $v_{\text{fusion}}$ is fed into a three-layer multilayer perceptron (MLP) with batch normalization, ReLU activations, and dropout regularization. The final output is a seven-dimensional logit vector:

$$z = \text{MLP}(v_{\text{fusion}}),$$

corresponding to the seven classes in the IDNet-EST dataset. A softmax layer converts logits into class probabilities during training.

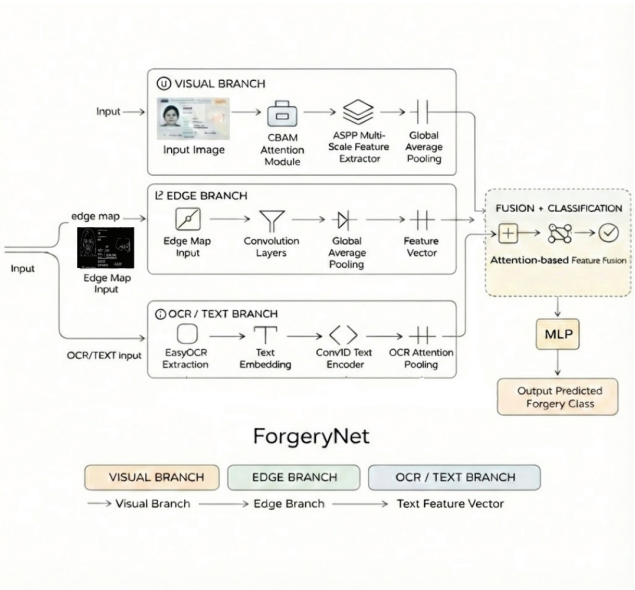



Fig. 3: System Architecture

### H. Training Strategy

The entire network is trained end-to-end using the AdamW optimizer with a learning rate of $1 \times 10^{-4}$. Cross-entropy loss is minimized to update model parameters. A ReduceLROnPlateau scheduler adaptively decreases the learning rate when validation performance stagnates, thereby improving convergence stability. Training is conducted for 25 epochs on a dataset of approximately 10,000 images, with best-model checkpointing based on validation performance to mitigate overfitting. Data augmentation and dropout regularization within the classifier further enhance generalization.

## V. Results and Discussion

### A. Classification Metrics

Table II presents the detailed classification performance of the proposed system across all forgery categories. The results demonstrate consistently high predictive accuracy, with precision, recall, and F1-scores exceeding 0.90 for most classes, indicating strong discriminative capability. Notably, the `face_replace` and `copy_move` categories achieve near-perfect precision and recall, reflecting the model's robustness in identifying visually subtle manipulations. The overall accuracy of 97.8% over more than 10,000 test samples further confirms the reliability and generalization strength of the proposed framework.

TABLE II: Classification Report

| Class | Precision | Recall | F1 | Support |
|---|---|---|---|---|
| Positive | 0.98 | 0.91 | 0.94 | 214 |
| Fraud1_Copy_Move | 1.00 | 1.00 | 1.00 | 213 |
| Fraud2_Face_Morph | 0.92 | 0.99 | 0.95 | 217 |
| Fraud3_Face_Replace | 1.00 | 1.00 | 1.00 | 206 |
| Fraud4_Combined | 1.00 | 1.00 | 1.00 | 211 |
| Fraud5_Inpaint_Rewrite | 0.91 | 0.99 | 0.95 | 215 |
| Fraud6_Crop_Replace | 0.98 | 0.90 | 0.94 | 214 |
| **Overall Accuracy** | | | **0.97** | 1490 |
| **Macro Average** | 0.97 | 0.97 | 0.97 | 1490 |
| **Weighted Average** | 0.97 | 0.97 | 0.97 | 1490 |

### B. Confusion Matrix

The raw confusion matrix shows that the majority of predictions are concentrated along the main diagonal, indicating a high number of correct classifications across all forgery categories. Classes such as positive, copy–move, face morph, face replace, combined, inpaint, and crop–replace exhibit strong diagonal dominance, reflecting robust true-positive performance. Limited misclassifications are observed primarily between visually similar categories, most notably from positive to combined and from combined to copy–move, suggesting ambiguity arising from overlapping manipulation artifacts.

### C. ROC and PR Curves

The ROC curves for all classes exhibit steep rises toward the upper-left region, yielding AUC values close to 1.0, which indicates excellent separability between authentic and forged samples. Likewise, the Precision–Recall (PR) curves remain consistently high across recall levels, demonstrating that the model maintains strong precision even under varying decision thresholds.


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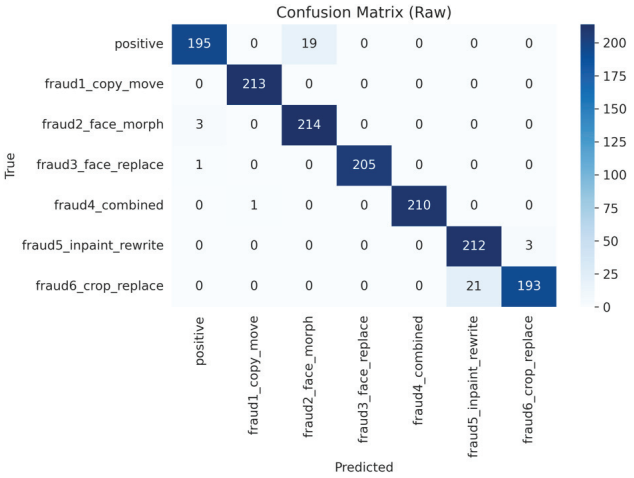



Fig. 4: Confusion Matrix

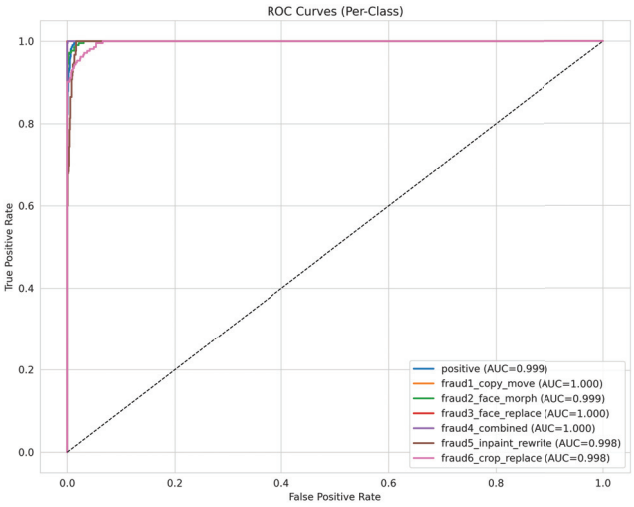



Fig. 5: ROC Curves

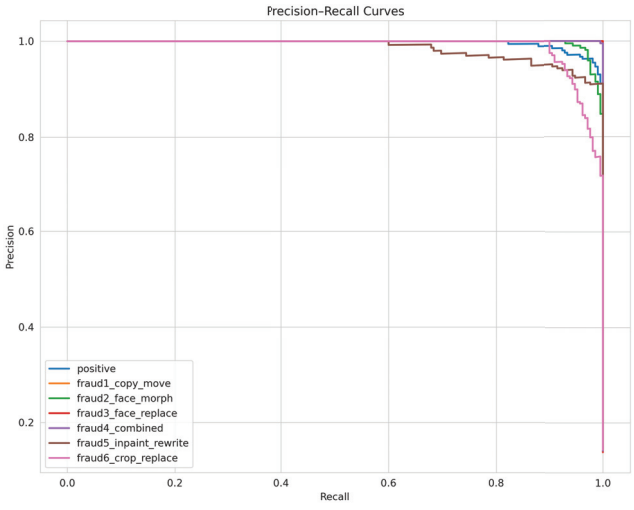



Fig. 6: Precision–Recall Curves

### D. Qualitative Analysis

Grad-CAM visualizations clearly highlight the manipulated regions within the identity document, including altered facial structures, tampered photo boundaries, and overwritten textual areas. The resulting heatmaps consistently concentrate on synthetic seams and forged patches, indicating that the model effectively attends to semantically meaningful forgery artifacts rather than irrelevant background content.

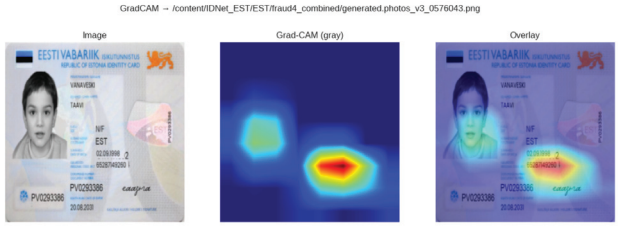



Fig. 7: Grad-CAM Analysis

## VI. Ablation Studies

To analyze the contribution of individual architectural components and input modalities, we conduct a detailed ablation study evaluating backbone selection, attention mechanisms, multi-scale feature extraction, and multi-branch fusion strategies.

TABLE III: Backbone Ablation Study

| Backbone | Val Acc (%) | Test Acc (%) | F1 |
|---|---|---|---|
| MobileNetV2 | **96.83** | **97.05** | **0.970** |
| EfficientNet-B3 | 94.13 | 93.36 | 0.933 |
| ResNet50 | 95.98 | 95.10 | 0.951 |

TABLE IV: Component-wise Ablation Study

| Model Variant | Val Acc (%) | Test Acc (%) | F1 |
|---|---|---|---|
| MobileNetV2 CNN | 97.82 | 97.72 | 0.9773 |
| MobileNetV2 + ASPP | 97.30 | 97.58 | 0.9768 |
| MobileNetV2 + CBAM | 97.69 | 98.26 | 0.9827 |
| Proposed (CBAM + ASPP) | **97.85** | **97.85** | **0.9787** |

TABLE V: Effect of Multi-Branch Feature Fusion

| Input Configuration | Val Acc (%) | Test Acc (%) | F1 |
|---|---|---|---|
| Image Only | 97.43 | 96.71 | 0.9671 |
| Image + OCR | 96.70 | 97.38 | 0.9739 |
| Image + Edge | 97.36 | 96.98 | 0.9699 |
| Full Multi-Branch (Proposed) | **97.82** | **97.85** | **0.9787** |



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TABLE VI: Final Performance of the Proposed Method

| Metric | Value |
|---|---|
| Validation Accuracy (%) | 97.82 |
| Test Accuracy (%) | 97.85 |
| Precision (Macro) | 0.9697 |
| Recall (Macro) | 0.9680 |
| F1-score (Macro) | 0.9787 |
| ROC-AUC (Macro) | 0.9992 |
| Matthews Corr. Coef. | 0.9627 |
| Cohen's Kappa | 0.9624 |

## VII. Conclusion

This work presents a multimodal deep-learning framework for reliable document forgery detection by integrating visual texture cues, structural edge patterns, and semantic inconsistencies extracted through OCR. The proposed architecture combines a MobileNet-based feature encoder with a dedicated edge-enhancement branch and a text-semantic verification module, enabling robust detection of print–scan recapture, region-level manipulation, font tampering, and textual rewriting.

Experimental results demonstrate that fusing low-level manipulation artifacts with high-level semantic context significantly improves generalization, particularly on challenging real-world forgeries. Overall, the proposed system offers a practical and scalable solution suitable for educational institutions, verification agencies, and government workflows requiring automated document integrity verification.

## VIII. FUTURE SCOPE

Although the proposed model demonstrates strong performance, several avenues remain for further enhancement:

- **Real-World Dataset Expansion:** Increasing the dataset size with diverse, high-resolution government and institutional document categories would improve robustness under varied printing, scanning, compression, and lighting conditions.
- **Transformer-Based Multi-Modal Fusion:** Future extensions may incorporate Vision Transformers (ViT), Document Transformers, or multimodal fusion architectures such as LayoutLMv3 or Donut to achieve more expressive cross-modal reasoning.
- **Fine-Grained Localization of Forged Regions:** Enhancing the model with pixel-level manipulation detection—via Grad-CAM++, SAM, or integrating a U-Net segmentation head—would enable explicit localization of forged or tampered areas.
- **Adversarial Robustness and Anti-Spoofing:** Incorporating adversarial training, noise-tolerant layers, or recapture-spoof detectors could strengthen the system against highly sophisticated AI-generated or GAN-based forgeries.
- **On-Device and Real-Time Deployment:** Optimization through model quantization, pruning, TensorRT acceleration, or ONNX export may support deployment on mobile devices, kiosks, and edge systems requiring real-time document verification.
- **Integration With Verification Pipelines:** Future iterations may integrate digital-signature verification, metadata consistency checks, or blockchain-based audit trails to form a more complete and secure document-verification ecosystem.

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